

neuroguessr-2: image geolocation

Technical report · 24 July 2026 · single static image, no panorama, no metadata · all numbers on the frozen 2 998-image validation set

Where we are. A DINOv3 ViT-L/16 classifier over 5 000 learned geocells, followed by a retrieval stage that snaps the answer onto the nearest visually-matching image among 1.2 M geo-tagged exemplars. Median error **43.75 km**, **42.9 %** of images within 25 km, **11.3 %** within 1 km, GeoGuessr equivalent **4 438 / 5 000** per round. For reference, PIGEON (Stanford, 2023) reports 131 km median and 24.2 % @25 km on single images.

1 · Task and evaluation

One 384×384 image in, one latitude/longitude out. Scoring is by great-circle (haversine) distance to the truth. The evaluator is **frozen**: the same 2 998 held-out images, the same metric code, never touched during research — so a score can be improved only by making the model better, never by making the test easier. Reported metrics:

- **median km** — the primary objective (robust to the hopeless tail)
- **acc@R** — share of images within R km, the fine-grained quality signal
- **mean km** — dominated by the ~2.5 % of images sent to the wrong continent; it is effectively a *country-accuracy* metric, not a fine-grained one
- **GeoGuessr score** — $5\,000 \cdot e^{-d/1492.7}$, the game's world-map formula

2 · Architecture

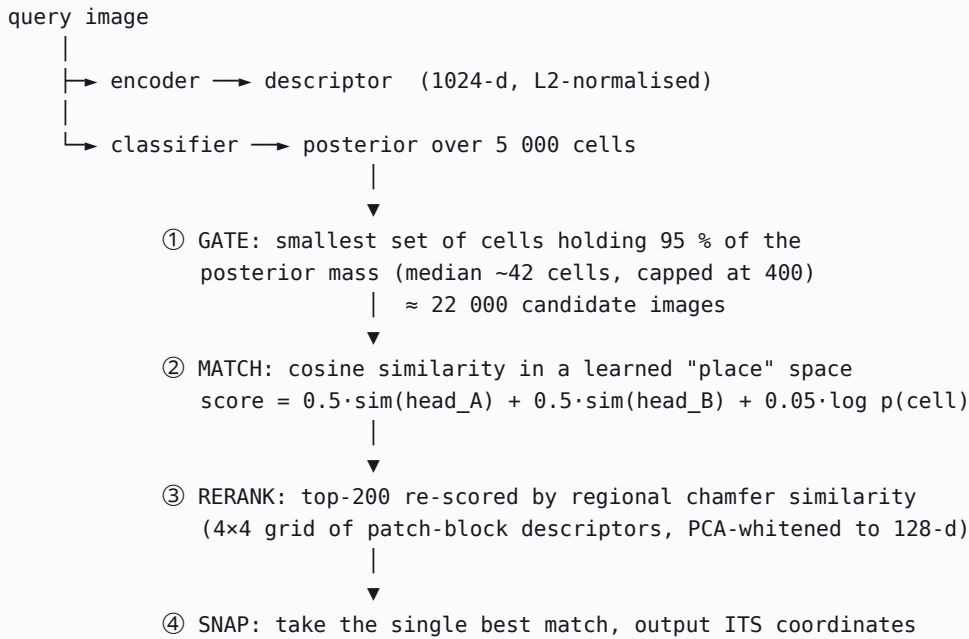
2.1 Stage one — the classifier (what it is)

component	detail	params
Backbone	DINOv3 ViT-L/16 (facebook/dinov3-vitl16-pretrain-lvd1689m), 384 px → 576 patch tokens + 5 prefix tokens, RoPE positions	~300 M frozen
Adaptation	LoRA rank 16 on every attention q, k, v, o projection	3.15 M
Trunk	LayerNorm → Linear(1024 → 1024) → GELU on the CLS token	1.05 M
Fine head A	5 000 geocells (k-means on training coordinates, country-constrained)	5.0 M
Fine head B	a second, <i>staggered</i> 5 000-cell tessellation (different seed) — two independent quantisations of the same globe, unioned at readout	5.0 M
Hierarchy	coarse heads at 64 and 512 cells, added to the fine logits through a child → parent map	0.6 M
Country head	log-softmax over countries, added via each cell's country	0.3 M
Total trainable (0.6 % of the backbone left frozen)		15.2 M

Training: 3 epochs over 1 198 072 images (299 518 locations × 4 headings) on 4×RTX 5090, cross-entropy on distance-smoothed cell targets (softmax(−d/τ), τ = 75 km) plus an auxiliary **mean-spherical-loss** term that back-propagates through the differentiable spherical mean of the predicted distribution with a Geman–McClure robust penalty in kilometres.

2.2 Stage two — retrieval (where the accuracy actually comes from)

The classifier can never beat its own cell size: with 5 000 cells the quantisation floor is ~34 km median, and the argmax readout can do no better. Retrieval removes that floor entirely by returning the coordinates of a *real training image* rather than a cell centre.



The index holds 1 198 072 descriptors (2.4 GB fp16 per space) plus $1.2 \text{ M} \times 16 \times 128$ regional descriptors (4.9 GB). One guess costs $\sim 1.2 \text{ s}$ on a laptop CPU.

2.3 The place heads — the decisive idea

Raw DINOv3 similarity answers "*do these two photos look alike*", which is the wrong question: two photos of the same 30 m of forest road taken 180° apart look nothing alike, while a Finnish road and a Canadian road look identical. Each head is a small residual MLP ($1024 \rightarrow 2048 \rightarrow 1024$, output layer zero-initialised so it starts as the identity) trained contrastively on *cached* descriptors — minutes of GPU time, no images touched.

The correction that mattered: the first head used same-location positives. But at test time the query's own location is never in the index — the best reachable answer is a *different* location 1–15 km away, and same-location training actively pushes those neighbours apart. Retraining with positives drawn from any location within 10 km, and masking in-batch negatives closer than 50 km as false negatives, moved the median from 61.2 $\rightarrow$ 49.0 km.

3 · What changed the numbers

step	what it was	median	@1 km	@25 km	mean
ratchet baseline	8-min experiments, 60 k-image subset	433	—	—	—
ratchet champion	~ 60 accepted/rejected experiments: geocells, hierarchy, country head, robust km loss, staggered tessellation	205	—	—	—
full run #1	same stack, all 1.2 M images, 3 epochs, 4 GPUs	104.7	0.0 %	7.6 %	456
full run #2	every banked win stacked at scale — <i>a wash</i> , establishing the 3-epoch regime ceiling	104.4	0.0 %	7.8 %	479
retrieval v1	mass gate + cosine kNN + snap, on frozen features	78.5	10.2 %	35.2 %	—
phase A	tuned gate (0.95), soft classifier prior ($\lambda = 0.05$), $k = 1$ snap	65.7	10.1 %	36.0 %	—
C1 place head	contrastive head on cached descriptors	61.2	11.5 %	38.7 %	390
C1.5 geo-smooth	positives within 10 km + near-negative masking; blend of two heads	49.0	11.6 %	41.2 %	388
C2 (current)	CLS $\oplus$ mean-patch descriptor blend + regional chamfer rerank	43.8	11.3 %	42.9 %	373

Everything from “retrieval v1” down is **inference-time only** — the classifier has not been retrained since run #2. Total compute cost of the entire retrieval program: ~\$6.

4 • Measured negatives (do not retry)

tried	result
512 px instead of 384 px (1.8× the compute)	+0.33 pts @25 km — resolution is <i>not</i> the lever, the most surprising result of the program
query-side test-time augmentation (flip, zoom)	−0.4 pts @25 km
multi-resolution ensemble (512 ⊕ 384, same model)	−0.9 pts — same model, correlated errors, no diversity to exploit
gate-mass sweep (0.90 / 0.95 / 0.99)	identical to 3 decimal places
consensus voting over top-k neighbours; location max-pooling; α-query-expansion	neutral to −0.5 pts — “trust the single nearest match” wins
post-trunk features instead of backbone CLS	worse everywhere — geo-classification training strips instance detail from the trunk

5 • Where the remaining error lives

Three ceilings, all measured on the same val set:

- **Coverage: 100 %.** For every single validation image there exists a training image within 25 km (median nearest 2.2 km). Data is not the constraint.
- **In-pool oracle: 92–98 %.** Even restricted to the candidates the gate actually shows, a perfect matcher would score 92–98 % @25 km.
- **We convert that into 42.9 %.**

So ~50 points of @25 km sit purely in **matching quality** — not in coverage, not in the gate, not in the cell scheme. Two rounds of head engineering moved it 38.7 → 42.9 with sharply diminishing returns, which is the evidence that the limit is in the *features themselves*: a backbone tuned to classify continents keeps what distinguishes Finland from Canada and discards what distinguishes one Finnish road from the next.

6 • Current distance profile

within	share	percentile	error
100 m	2.97 %	p10	0.8 km
500 m	7.74 %	p25	5.4 km
1 km	11.27 %	p50	43.8 km
5 km	24.15 %	p75	211 km
10 km	31.72 %	p90	564 km
25 km	42.93 %	p99	7 986 km
50 km	51.83 %		
200 km	73.92 %	GeoGuessr	score
1 000 km	94.50 %	mean per round	4 438
		median round	4 856
		5-round game	22 188
		rounds ≥ 4 500	69.3 %

7 • In flight

C3 — contrastive fine-tune of the encoder itself. Every head so far could only re-weight information the frozen backbone already kept. C3 trains the backbone (LoRA) so that cosine similarity *is* geographic proximity: positives within 10 km, negatives drawn from the same ~500 km region so they are visually similar but geographically wrong, near-pairs masked out of the loss, embeddings gathered across all four GPUs for a large negative pool. The classifier is deliberately untouched — the retrieval gate keeps using the old checkpoint's logits, so a worse encoder cannot poison the gate. Expected @25 km: 46–49 %.

Source: `train_full.py` (classifier), `retrieval/` (embed_c2, train_geo_head, train_c3_contrastive, eval_c2_gpu, engine), full experiment ledger in `RESEARCH_LOG.md`.